

Project Demonstration

Date	4 July 2026
TeamID	XXXXXX
ProjectName	EduGenie – AI-Powered Educational Assistant
MaximumMarks	5 Marks

Project Demonstration Overview

This section demonstrates the complete workflow of EduGenie – AI-Powered Educational Assistant. The application provides intelligent educational support by answering academic questions, explaining complex concepts in simple language, generating topic-based quizzes, summarizing educational content, and creating personalized learning paths. The system integrates FastAPI, HTML/CSS, and Generative AI models to deliver a responsive and interactive learning experience. The demonstration verifies that all core modules operate successfully and satisfy the project requirements.

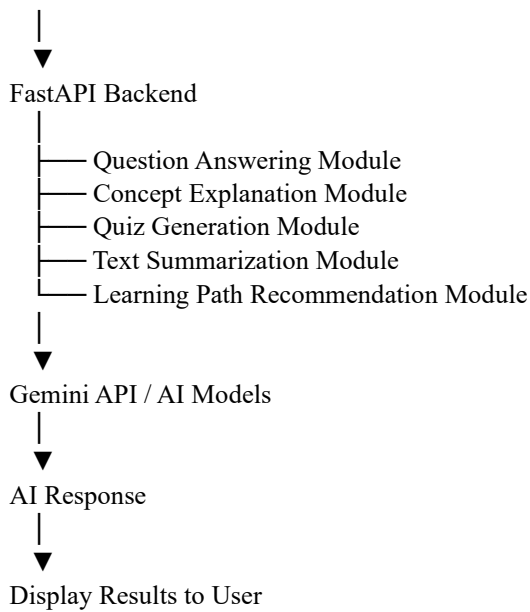
1. System Architecture

The architecture of EduGenie consists of the following components:

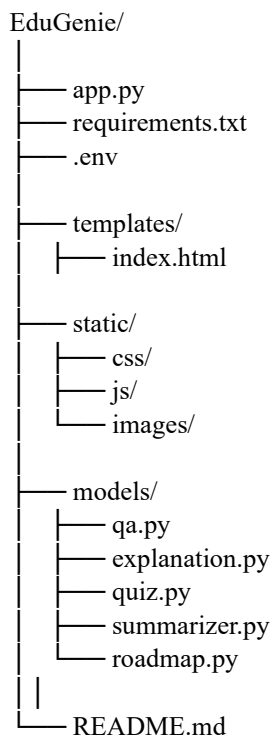
- Frontend: HTML, CSS, and Jinja2 templates provide an interactive and user-friendly interface.
- Backend: FastAPI handles HTTP requests, API routing, and communication between modules.
- AI Models: Gemini API and LaMini-Flan-T5 process educational queries and generate intelligent responses.
- Utility Modules: Manage configuration, validation, and error handling.
- Environment Configuration: API keys and application settings are securely managed using the .env file.

Workflow:





2. Project Structure



3. Running the Application

Prerequisites

- Python 3.10 or above
- FastAPI
- Uvicorn
- Gemini API Key

- Virtual Environment

Install Dependencies

`pip install -r requirements.txt`

Run the Application

`uvicorn app --reload`

Open the Application <http://127.0.0.1:8000>

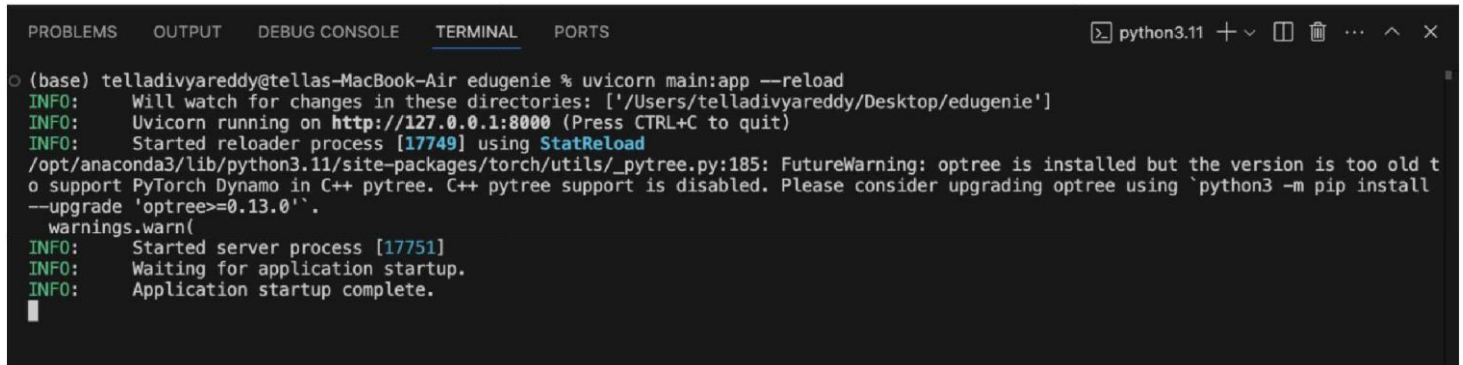
A terminal window with a dark background and light green text. The terminal shows the command `uvicorn main:app --reload` being executed. The output includes several informational messages: `INFO: Will watch for changes in these directories: ['/Users/telladivyaireddy/Desktop/edugenie']`, `INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)`, `INFO: Started reloader process [17749] using StatReload`, a `FutureWarning` about `optree` being outdated, `INFO: Started server process [17751]`, `INFO: Waiting for application startup.`, and `INFO: Application startup complete.`. The terminal window has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL (selected), and PORTS. The title bar shows 'python3.11' and standard window controls.

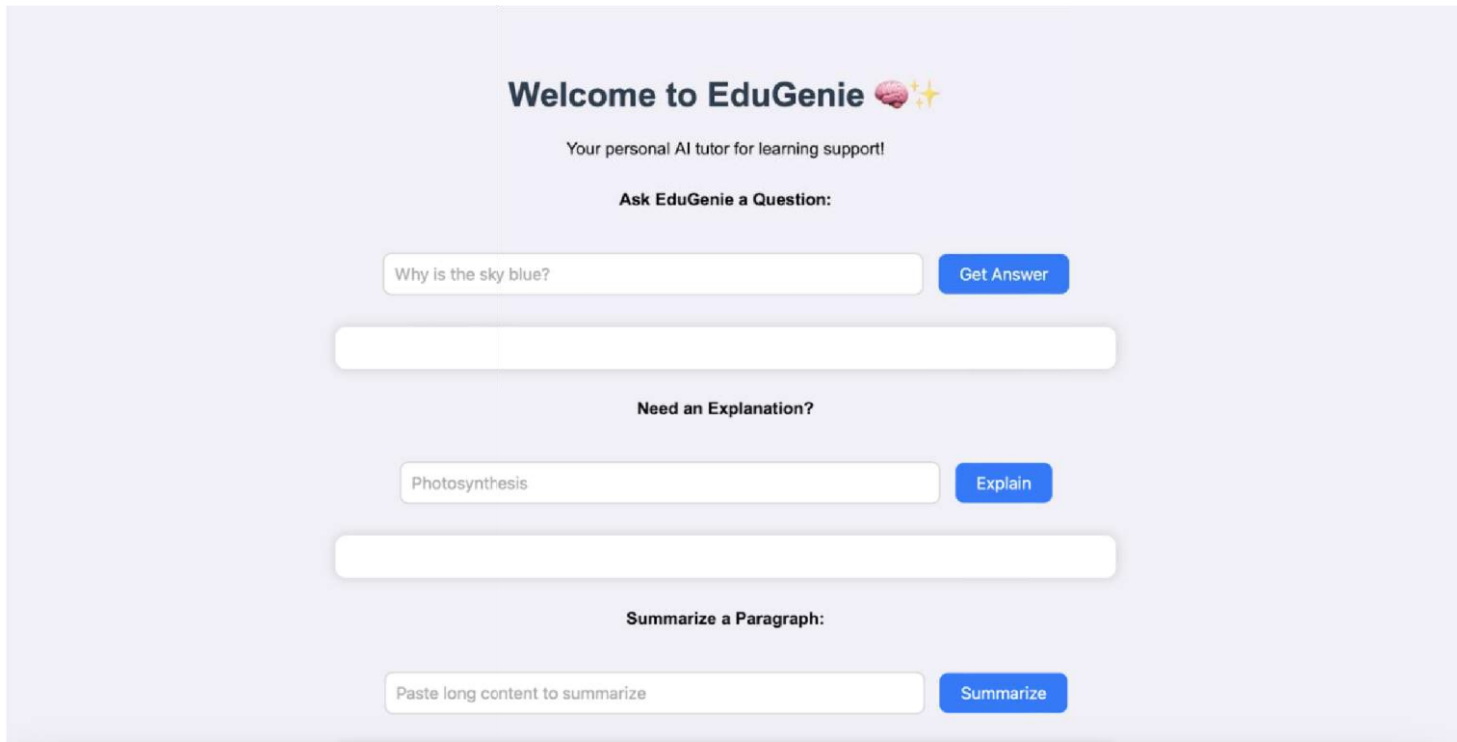
Fig. Showing the Application can be accessed

The FastAPI server starts successfully, and the EduGenie homepage is displayed in the web browser.

4. Application Running

The application launches successfully and provides a responsive interface where users can select different educational AI services. Navigation between modules is seamless, and responses are displayed in real time.

5. Application Workflow

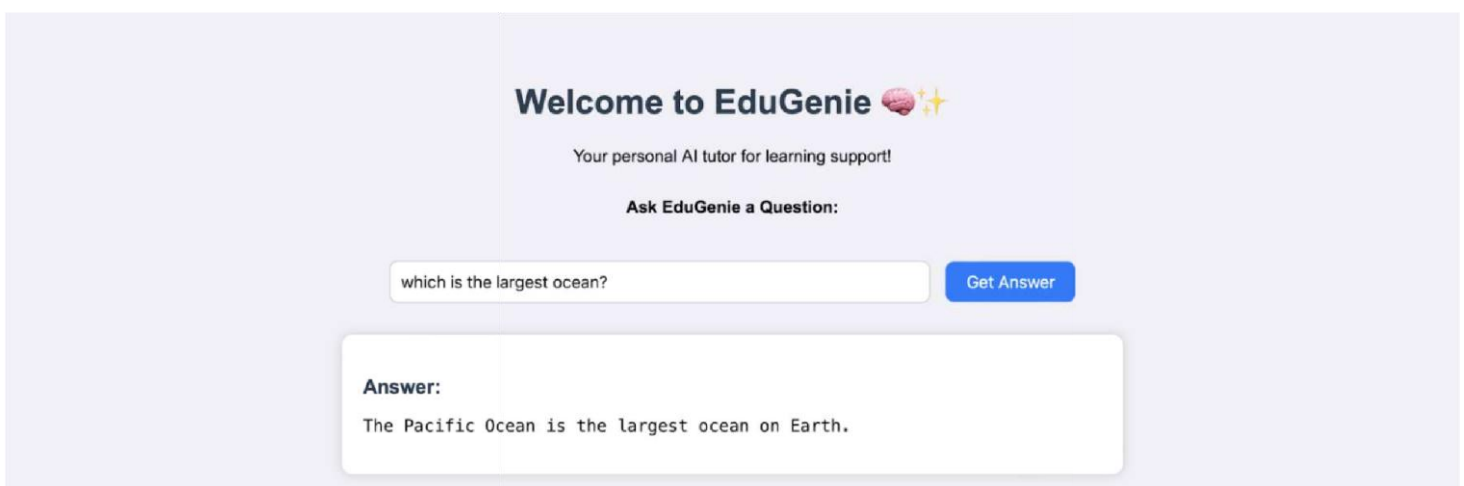


The image shows the EduGenie Home Page interface. At the top, it says "Welcome to EduGenie" with a brain and star icon, followed by "Your personal AI tutor for learning support!". Below this is a section titled "Ask EduGenie a Question:" with a text input field containing "Why is the sky blue?" and a blue "Get Answer" button. A large white box is positioned below the input field. The next section is titled "Need an Explanation?" with a text input field containing "Photosynthesis" and a blue "Explain" button. Another large white box is below this. The final section is titled "Summarize a Paragraph:" with a text input field containing "Paste long content to summarize" and a blue "Summarize" button.

Figure 1: EduGenie Home Page

Displays the main interface with options for:

- Intelligent Question Answering
- Concept Explanation
- Quiz Generation
- Text Summarization
- Learning Path Recommendation



The image shows the EduGenie Intelligent Question Answering interface. It features the same header as Figure 1. Below the "Ask EduGenie a Question:" section, the text input field now contains "which is the largest ocean?" and the "Get Answer" button is visible. Below the input field, a large white box displays the answer: "Answer: The Pacific Ocean is the largest ocean on Earth."

Figure 2: Intelligent Question Answering

The user enters an educational question, and EduGenie generates an accurate answer with additional learning context.

Need an Explanation?

Binary Search Algorithm

Explain

Explanation:

The binary search algorithm is a way to find the index of a target value in a sorted list of numbers. It's like searching for a needle in a haystack. You need to find the number between 0 and 1, and then go back to the beginning of the list to find the target value.

Figure 3: Concept Explanation

Users submit a topic or concept, and the system provides a simplified explanation suitable for learners.

Generate a Quiz:

Pythagoras theorem

Generate Quiz

Quiz:

Q1: What does the Pythagorean theorem describe?

☐ The relationship between the angles of a triangle.

☐ The relationship between the sides of a right-angled triangle.

☐ The relationship between the area and perimeter of a triangle.

☐ The relationship between the sides of any triangle.

Check Answer

Figure 4: AI Quiz Generation

The application generates topic-specific multiple-choice questions for self-assessment.

Summarize a Paragraph:

The Industrial Revolution, which began in the late 18th century, mark

Summarize

Summary:

The Industrial Revolution changed the world by shifting it from farming to factories. New machines, like James Watt's steam engine, made things faster and easier to produce. While this boosted economies, it also created problems like bad working conditions, child labor, pollution, and overcrowded cities. Even with these problems, the Industrial Revolution shaped the modern world, affecting how we travel, communicate, work, and trade.

Figure 5: Educational Text Summarization

Long educational content is summarized into concise and easy-to-understand key points.

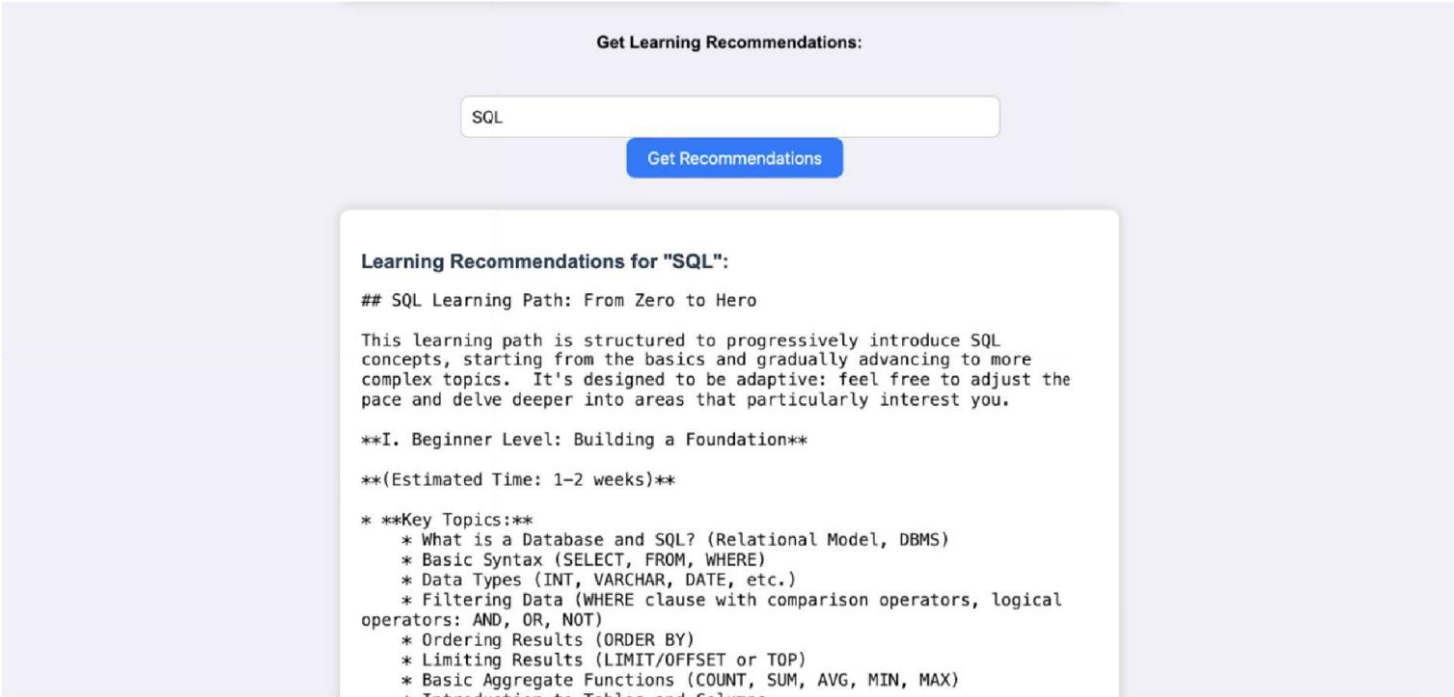


Figure 6: Personalized Learning Path

The system creates a structured learning roadmap covering beginner, intermediate, and advanced topics with recommended study progression.